

Corey Zhou

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PROFESSIONAL SUMMARY

Computer engineering graduate with hands-on RTL-level verification experience, including constrained-random testing, SystemVerilog assertions, and functional coverage on a RISC-V processor pipeline. Completed UVM certification with applied project experience building a full UVM environment from scratch. Strong debugging fundamentals and a track record of taking ownership of complex technical problems.

SKILLS

Hardware Design & Verification: SystemVerilog, UVM (coursework/self-study), Constrained-Random Verification, SystemVerilog Assertions (SVA), Functional Coverage, Verilog, MIPS32 Assembly, FPGA, ASIC/SoC, PCB

EDA / Simulation Tools: ModelSim, Questa, HSPICE, Vivado, Autodesk EAGLE, KiCAD

Programming Languages: C/C++, Python, Java, JavaScript/TypeScript

Scripting & Test Automation: Pytest, Selenium, Jest

Systems: Linux/Unix, Windows

Additional: Docker, Jenkins, Git, CI/CD, AWS, Terraform, Networking (TCP/IP, ARP)

EDUCATION

University of California, Santa Barbara

B.S. Computer Engineering. Dean's Honors. GPA: 3.78.

Relevant Coursework: Verification Series (Udemy Certificate, 2026): SystemVerilog, UVM, Assertions, Functional Coverage

PROJECTS

RISC-V Processor Functional Verification

- Built a SystemVerilog verification environment with constrained-random assembly test programs and SVA checks, covering 11 instructions across 5 randomized test runs per instruction to detect functional bugs in a RISC-V processor pipeline.
- Debugged simulation failures using waveform analysis, tracing root causes through the pipeline and identifying 8 functional bugs across the instruction set.

Dual-Port RAM UVM Verification Environment

- Built a complete UVM verification environment (driver, monitor, agent, scoreboard, sequence, subscriber) to verify a synchronous dual-port RAM with independent read and write ports.
- Wrote sequences generating randomized, pipelined transactions exercising concurrent read/write behavior; built a scoreboard with a reference memory model to flag mismatches on every read.

5-Stage 32-bit Pipelined MIPS32 Processor

- Designed and verified a complete 5-stage pipelined MIPS32 processor (fetch, decode, execute, memory, writeback) in Verilog, with a hazard unit for data forwarding and stall detection.
- Built a 2-bit, 2-way branch predictor, a 32KB 2-way set-associative cache (4-word blocks), and a combinational multiplier for multiply/divide, each with a dedicated testbench.

Full-Custom 16-bit Carry Skip Adder

- Designed a full-custom 16-bit carry skip adder using pass transistor logic, sense amplifiers, and complementary domino logic to generate PGK sum signals at full voltage swing.
- Simulated every custom cell and full adder in HSpice, characterizing delay and power across PVT corners to validate design margins.

EXPERIENCE

Founding Engineer

October 2025 – Present

SlamOptix

- Joined as founding engineer with full ownership of architecture and technical direction, building core products end-to-end as the sole engineer (MongoDB, Node.js, Next.js, NGINX).
- Built an AI-assisted pipeline (self-hosted LLM inference, automated test/integration) that reduced manual effort on routine tasks.

Math Interactives Intern

July 2018 – September 2019

CK-12 Foundation

Palo Alto, CA

- Designed interactive math content for middle and high school students and reviewed peers' work using GeoGebra's JavaScript API.